Estela Pérez Luque, PhD

Ergonomics Engineer – Automotive

+46 73 262 8876 • perezluque.estela1504@gmail.com • LinkedIn • Skövde, Sweden

PROFILE

Automotive ergonomics engineer specializing in occupant packaging and human-centered vehicle design. I translate user needs, anthropometric variability, and real-world data into structured evaluation, validation, and design recommendations for vehicle interiors. My work combines digital human modelling, simulation, and experimental data to assess visibility, reachability, driver position, entry and exit, and comfort in early development phases. I build reproducible evaluation workflows and collaborate with cross-functional teams to support decision-making in complex engineering environments.

TECHNICAL SKILLS

Core tools: Python, MATLAB, IPS IMMA, Siemens Jack, AnyBody, Autodesk Inventor, SolidWorks, NX

Methods: V&V, simulation-based evaluation, benchmarking, end-to-end data analysis, time-series data, motion capture

Domain: Human factors, automotive ergonomics, occupant packaging, experimental data systems.

Languages: English (C1), Spanish (Native), Swedish (B1-improving).

PROFESSIONAL EXPERIENCE

Ergonomics Researcher, University of Skövde (Sweden) — **Sep 2020 – Oct 2025**

- Evaluated 2 vehicle configurations and 199+ driver cases to analyse H-point, eye-point, and steering wheel positioning, supporting ergonomic decision-making for visibility, reachability, and comfort.
- Designed and executed 10+ scenario-based validation studies, comparing simulation outputs with motion capture and in-vehicle datasets, generating traceable evidence for model assessment and design decisions.
- Built reproducible Python and MATLAB pipelines to benchmark model performance across participants, tasks, and conditions, reducing manual analysis effort and enabling consistent validation workflows.
- Developed quantitative evaluation methods (RMSE, DTW, multivariate metrics) to assess agreement between simulated and observed motion, improving the robustness of validation processes.
- Delivered decision-ready recommendations to industry partners (Volvo Cars, Volvo Trucks, Scania), translating complex analysis into actionable inputs for vehicle design and ergonomics evaluation.
- Contributed to 15+ peer-reviewed publications, strengthening credibility and dissemination of validated methods within automotive ergonomics and simulation..

Visiting Researcher - University of Antwerp (Belgium) — **Sep 2024 – Mar 2025**

- Planned and executed multi-site experimental studies combining motion capture and 4D body scanning, ensuring consistency across institutions.
- Developed end-to-end data pipelines for large-scale motion and body datasets (180+ frames per sequence), enabling reproducible analysis and validation.
- Delivered standardized workflows and documentation used by multiple research teams to support data-driven ergonomics evaluation.

Co-founder and R&D Lead - Swe-Ex AB (Startup) (Sweden) — **Sep 2021 – Mar 2023**

Estela Pérez Luque, PhD

Ergonomics Engineer – Automotive

+46 73 262 8876 • perezluque.estela1504@gmail.com • LinkedIn • Skövde, Sweden

- Co-founded a startup developing wearable ergonomic support equipment for assembly-line environments, targeting injury prevention and worker safety.
- Led development of 3 functional prototypes, integrating CAD design, material selection, and user testing under real-world constraints.
- Translated user needs and ergonomic requirements into measurable design criteria and product specifications, guiding prototype iterations toward improved usability and safety.
- Conducted prototype validation tests and stakeholder demos, supporting technical feasibility and product direction decisions.
- Developed business and product perspective through a 2-year incubator program, working on value proposition, prioritization, and feasibility under uncertainty.

Research Assistant - University of Skövde (Sweden) **Jun 2019 – Sep 2020**

- Conducted ergonomic assessments (RULA, REBA) for 6+ industry partners, identifying high-risk postures and recommending design improvements.
- Delivered practical reports and visual documentation to support product development discussions and usability improvements..

EDUCATION

PhD in Informatics, University of Skövde (Sweden) Sep 2020 – Oct 2025

PhD Thesis: Human Posture and Motion Prediction for Automotive Ergonomics Design.

Focused on simulation accuracy, anthropometric variability, and integration of ergonomics into early vehicle development. Developed and validated posture and motion prediction methods for occupant packaging using motion capture data and digital human modeling.

Master of Science in Intelligent Automation, University of Skövde (Sweden) Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden) Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain) Aug 2012 – Jun 2018

RESEARCH PUBLICATIONS

Full list available at: Google Scholar – Estela Pérez Luque